Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHNOLOGY

(Autonomous Institute Affiliated to University of Mumbai)
Munshi Nagar, Andheri (W), Mumbai – 400 058.

Mid Semester Examination
Apr - 2025

Max. Marks: 30 | **Duration: 60 Min.**
Class: TE(CE/CSE) | **Semester: VI**
Course Code: PC1X
Name of the Course: Software Engineering

Instruction:

1) **All questions are compulsory.**
2) **Draw neat diagrams.**
3) **Assume suitable data if necessary but justify the same and interpret ambiguous text. Solve questions in order Q1, Q2, Q3. Start new question on a new page.**

Course Outcomes (CO): At the End of the course students will be able to

1X.1 Analyze software requirements.
1X.2 Apply UML models for a project.
1X.3 Evaluate system architecture and develop a detailed task schedule from the overall estimates and planning.
1X.4 Illustrate different coding principles with unit test process.
1X.5 Understand the need for DevOps.

Q. No.	Question	Max. Marks	CO-BL
Q. 1	You are appointed as a software engineer for implementing the Online examination paper evaluation system of a college. Design the class diagram and use case diagram for the same. The class diagram should demonstrate inheritance, class member visibility, and cardinality. The use case diagram should include 5 use cases (add includes and extends wherever needed). Further, explain one of the use cases in detail and include scope, brief, pre and post conditions, trigger, basic flow and extensions.	10	CO-1, BL-6
Q. 2	As the software designer for a movie theatre ticket booking system - illustrate the data flow diagram level 0 and 1 and also the sequence / interaction diagram. The data flow diagram should include multiple stakeholders. The interaction diagram should include alternatives, options, loops, messages, lifelines, guards, system boundary, name tag, break, parallel, references to outside sequence diagrams. Use Gane and Sarson symbols for the data flow diagrams. DRAW LOGICAL DFD.	10	CO-2, BL-4
Q. 3	Create a deliverable based work breakdown structure, Gantt and Program evaluation review technique for implementing a library management system. Work breakdown structure should have three levels with work percentage and budget in Rupees. Next integrate WBS into a Gantt chart and create a PERT chart with a network diagram with earliest start time and latest completion time calculations. CALCULATE CRITICAL PATH AND COST.	10	CO-3, BL-5